

Name: 黃若綺

ID: B14208015

Department: 地理系

Reference : Stewart §4.1, 4,7

Minimizing Costs.

Let $C(x)$ be the cost of producing x units of a commodity. Part of the cost is independent of the output level x and is referred as the **fixed cost**, whereas the rest of the cost that depends on x is called the **variable cost**. In other words, if we denote the fixed cost by F and the variable cost by $C_v(x)$, then

$$C(x) = F + C_v(x)$$

and by convention we assume that $F = C(0)$. We further define :

- the *average cost function* by $AC(x) = \frac{C(x)}{x}$,
- the *average variable cost function* by $AVC(x) = \frac{C_v(x)}{x}$,
- the *marginal cost function* by $MC(x) = \frac{d}{dx}C(x)$.

The goal of a firm is to minimize the average cost or the average variable cost.

Question 1. Suppose that the cost function is given by $C(x) = x^3 - 9x^2 + 6x + 25$.

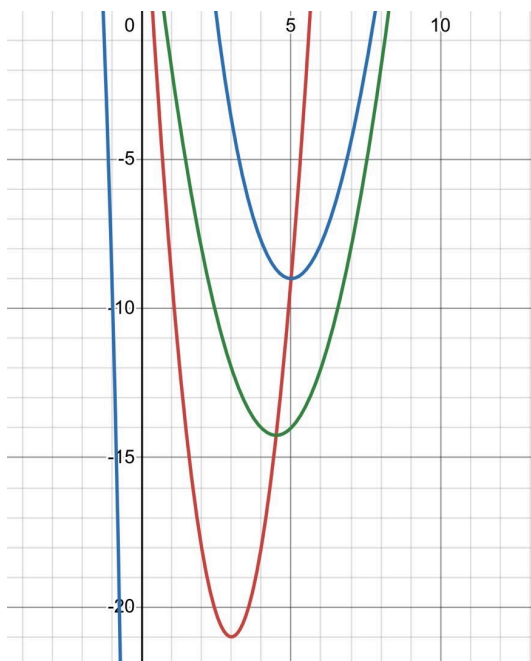
- (a) Derive the marginal cost function $MC(x)$, the average cost function $AC(x)$, and the average variable cost function $AVC(x)$. Sketch the graphs of these functions on the same figure.

$$C(x) = x^3 - 9x^2 + 6x + 25, \quad x \geq 0$$

$$MC(x) = C'(x) = 3x^2 - 18x + 6$$

$$AC(x) = \frac{C(x)}{x} = \frac{x^3 - 9x^2 + 6x + 25}{x} = x^2 - 9x + 6 + \frac{25}{x}, \quad x > 0$$

$$AVC(x) = \frac{x^3 - 9x^2 + 6x}{x} = x^2 - 9x + 6, \quad x > 0$$



$MC(x)$ —
 $AC(x)$ —
 $AVC(x)$ —

- (b) Find the output level $x > 0$ such that (i) the average cost is minimized,
(ii) the marginal cost function intersects the average cost function.

Are these answers the same?

$$i) AC(x) = x^2 - 9x + 6 + 25x^{-1}, x > 0$$

$$AC'(x) = 2x - 9 - 25x^{-2} \Rightarrow 2x - 9 - 25x^{-2} = 0 \Rightarrow x = 5$$

$$AC'(x) > 0, \text{ if } x > 5$$

$$AC'(x) < 0, \text{ if } x < 5$$

$\Rightarrow AC(x)$ has minimum at $x = 5$

$$ii) 3x^2 - 18x + 6 = x^2 - 9x + 6 + 25x^{-1}, x > 0$$

$$2x^2 - 9x - 25x^{-1} = 0 \Rightarrow x = 5$$

i) and ii) have the same answer

- (c) Find the output level $x > 0$ such that (i) the average variable cost is minimized,
(ii) the marginal cost function intersects the average variable cost function.

Are these answers the same?

$$i) AVC(x) = \frac{x^2 - 9x + 6x}{x} = x^2 - 9x + 6, x > 0$$

$$AVC'(x) = 0 \Rightarrow x = \frac{9}{2}, AVC\left(\frac{9}{2}\right) = \frac{81}{4} - \frac{9}{2} + 6 = \frac{87}{4}$$

when $x = \frac{9}{2}$, AVC has minimum $\frac{87}{4}$

$$ii) 3x^2 - 18x + 6 = x^2 - 9x + 6, x > 0$$

$$2x^2 - 9x = 0 \Rightarrow x = 0 \vee \frac{9}{2} \text{ and } x > 0 \Rightarrow x = \frac{9}{2}$$

Question 2. In general, consider any differentiable cost function.

- (a) Prove that the marginal cost function always passes through the minimum points of both the average cost function and the average variable cost function.

ⓐ For $x > 0$, if the $AC(x)$ obtains minimum value at $x > 0$
 then by Fermat's Theorem, $AC'(x) = \frac{x \cdot C'(x) - C(x)}{x^2} = 0$
 $\Rightarrow x \cdot C'(x) - C(x) = 0 \Rightarrow x C'(x) = C(x)$
 $\Rightarrow C'(x) = \frac{C(x)}{x} \Rightarrow MC(x) = AC(x)$

ⓑ if the $AVC(x)$ obtains minimum value at $x > 0$
 then by Fermat's Theorem, $AVC'(x) = \frac{x(C'(x) - C'(0)) - C(x) + C(0)}{x^2} = 0$

- (b) Show that $\lim_{x \rightarrow 0^+} AVC(x) = MC(0)$.

$$\lim_{x \rightarrow 0^+} AVC(x) = \lim_{x \rightarrow 0^+} \frac{C(x) - F}{x} = \lim_{x \rightarrow 0^+} \frac{C(x) - C(0)}{x - 0} = C'(0) = MC(0)$$

Maximizing Profits.

Now we look at the *profit* of a firm which equals to the revenue minus the cost. i.e. if $R(x)$ denotes the revenue when x units are produced and sold and $C(x)$ represents the cost, then the profit of selling x units equals to $\Pi(x) = R(x) - C(x)$. It is assumed that the second derivatives of both $R(x)$ and $C(x)$ exist and are continuous.

Usually, the goal of a firm is to maximize the profit. We will discuss some interesting consequences when profit is maximized by using the First and Second Derivative Tests.

Question 3.

- (a) Prove that if the profit attains a local maximum at $x_0 > 0$, then $R'(x_0) = C'(x_0)$ and $R''(x_0) \leq C''(x_0)$.

⊙ $R(x)$ and $C(x)$ are differentiable at $(0, \infty)$
 if $\Pi(x) = R(x) - C(x)$ obtains maximum value at $x_0 > 0$
 then $\Pi(x)$ is a local maximum by the Fermat's Theorem
 $\Pi'(x_0) = R'(x_0) - C'(x_0) = 0 \Rightarrow R'(x_0) = C'(x_0)$
 ⊙ Suppose that $R'(x_0) > C'(x_0)$, Then $\Pi'(x_0) > 0$
 and $\Pi(x_0)$ is a local minimum, contradicts to the fact $\Rightarrow$ so, $R''(x_0) \leq C''(x_0)$

- (b) In a *competitive* market, the unit price of a commodity is a constant p_0 . Write down, in this case, the revenue function $R(x)$. Deduce that when the profit is maximized, the marginal cost equals to p_0 .

$R(x) = p_0 \cdot x$
 when the profit is the maximum at $x = x_0 > 0$
 $R'(x_0) = C'(x_0)$
 However $R'(x_0) = p_0$
 $\Rightarrow C'(x_0) = p_0 = MC(x_0)$

- (c) Suppose that for a certain commodity,

- when x units are demanded, the unit price is $p(x) = -0.018x + 15$,
- the cost of producing x units is $C(x) = 1500 - 3x + 0.012x^2$.

Write down its revenue function $R(x)$ and its profit function $\Pi(x)$. Then find the maximum profit.

$R(x) = p(x) \cdot x = -0.018x^2 + 15x, x \geq 0$
 $\Pi(x) = R(x) - C(x) = -0.018x^2 + 15x - 1500 + 3x - 0.012x^2 = -0.03x^2 + 18x - 1500, x \geq 0$
 $\Pi'(x) = -0.06x + 18, x \geq 0 \Rightarrow x = 300$
 $\Pi(300) = 2200$

$\Pi'(x) < 0$, if $x > 300$
 $\Pi'(x) > 0$, if $0 \leq x < 300$
 $\Pi(x)$ has the maximum 2200 at $x = 300$